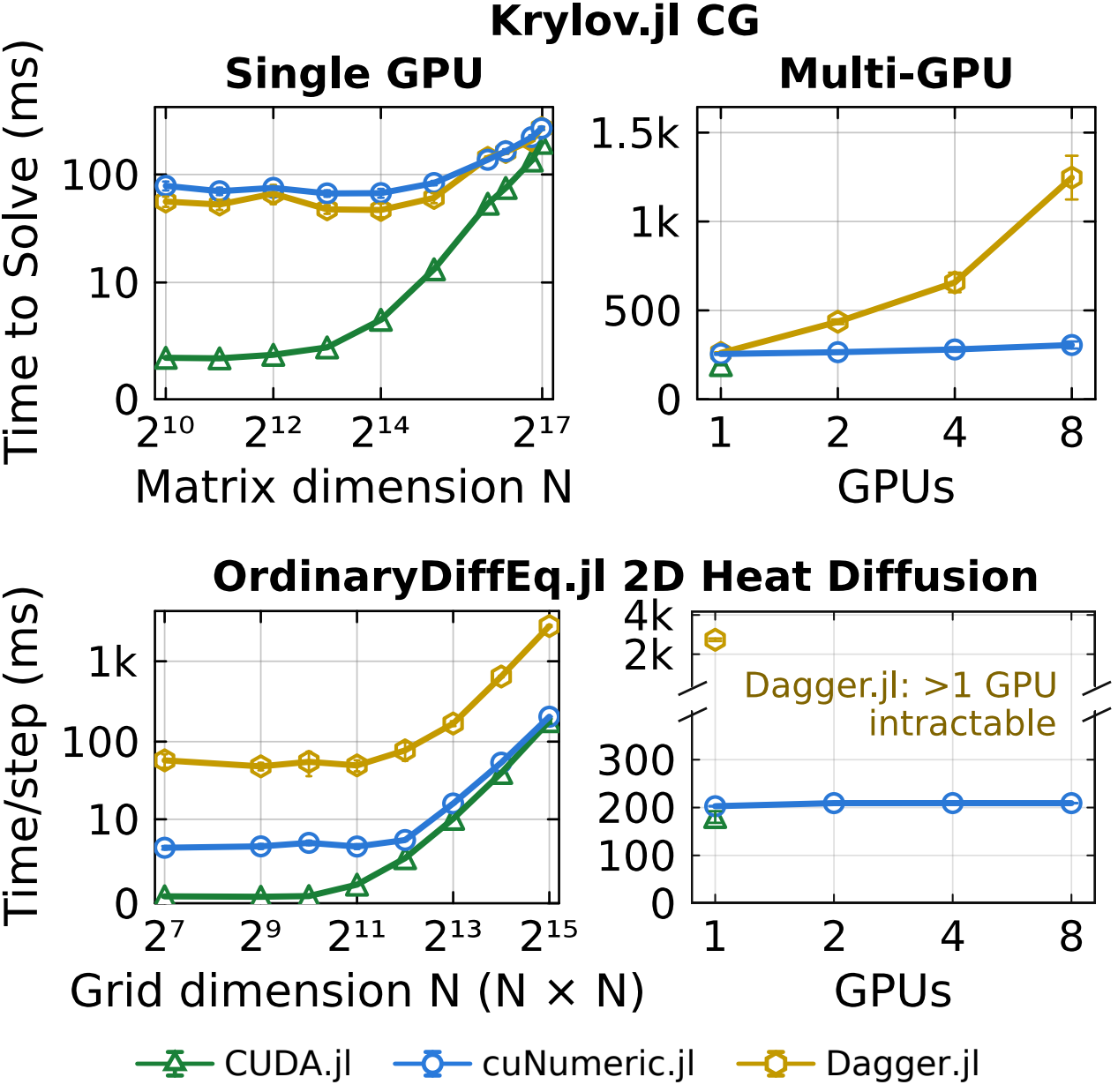

Krylov.jl CG
Single GPU
Multi-GPU
Time to Solve (ms)
100
10
0
$2^{10}$
$2^{12}$
$2^{14}$
$2^{17}$
Matrix dimension N
1.5k
1k
500
0
1
2
4
8
GPUs
OrdinaryDiffEq.jl 2D Heat Diffusion
Time/step (ms)
1k
100
10
0
$2^{7}$
$2^{9}$
$2^{11}$
$2^{13}$
$2^{15}$
Grid dimension N (N × N)
4k
2k
Dagger.jl: >1 GPU
intractable
300
200
100
0
1
2
4
8
GPUs
CUDA.jl
cuNumeric.jl
Dagger.jl